

1. *These are open questions. So it is fine as long as the arguments are reasonable. The following answers are for reference.

In the following discussions, we let kids' ability *abil* represent the error term u .

For (1), the random assignment of kids implies that *STR* and *abil* will be independent from each other. In this case, $E[abil|STR] = E[abil]$. Furthermore, we have $E[abil] = 0$ as we can always assume $E[u] = 0$, where $E[u]$ denotes the unconditional mean of the error term u . Therefore, OLS Assumption 1 will hold in this case.

For (2), if good schools tend to have low *STR* (as they have more resources to hire more teachers), while bad schools tend to have high *STR*, the selection with regard to kids' abilities implies that *abil* and *STR* will be negatively correlated (on average *abil* is low when *STR* is high, and vice versa). Therefore, OLS Assumption 1 will be violated in this case.

For (3), it is reasonable to argue that all parents will first try the good schools (assuming they all care about their kids). In this case, a random balloting will occur as the number of applicants will exceed the vacancy of the good schools. As a consequence of the random balloting, kids' with different levels of ability will have the same chances to be enrolled into good schools and bad schools, so that OLS Assumption 1 is likely to hold.

2. (a) R^2 measures the fraction of the sample variation in Y that is explained by X . Notice that Y_i can be decomposed as

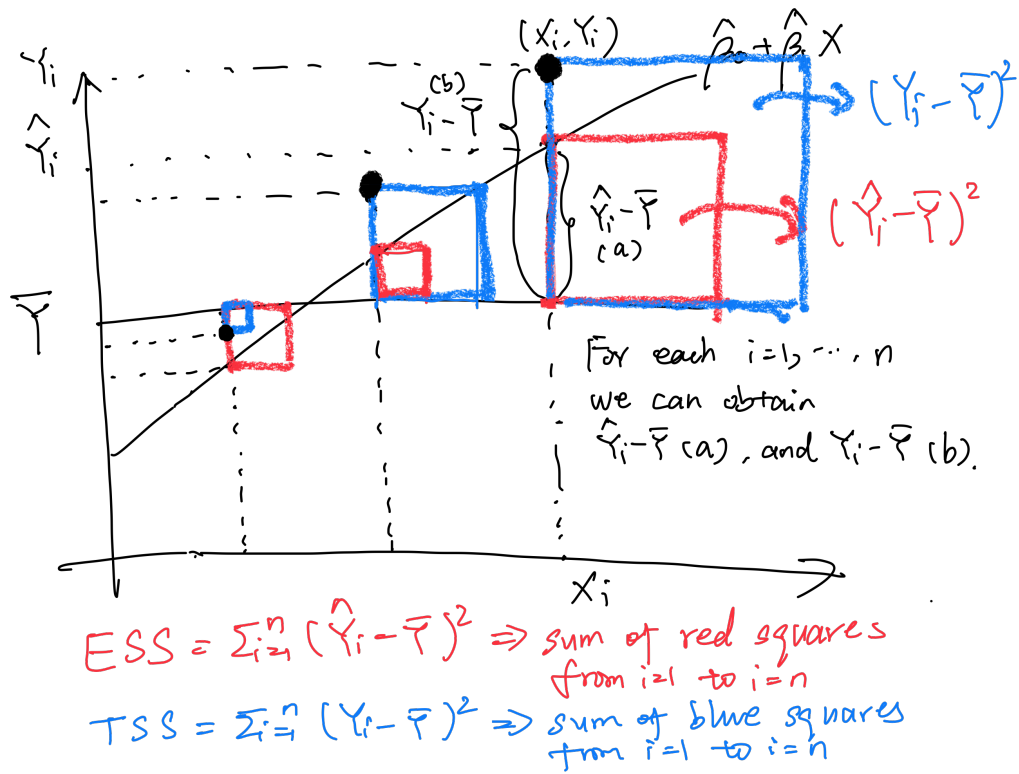
$$Y_i = \hat{Y}_i + \hat{u}_i,$$

where $\hat{Y}_i$ is the OLS predicted value and $\hat{u}_i$ is the residual. R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y .

Specifically, the R^2 can be written as the ratio of the explained sum of squares (ESS) to the total sum of squares (TSS):

$$R^2 = \frac{ESS}{TSS}, \quad ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2, \quad \text{and} \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

- (b) ESS is equal to the sum over $i = 1, \dots, n$ of the square of the difference between $\hat{Y}_i$ and $\bar{Y}$ ($\hat{Y}_i - \bar{Y}$, part (a) in the following graph), and TSS is equal to the sum over $i = 1, \dots, n$ of the square of the difference between Y_i and $\bar{Y}$ ($Y_i - \bar{Y}$, part (b) in the following graph). The difference between Y_i and $\hat{Y}_i$ is the residual $\hat{u}_i$. Furthermore, although for some i , $(\hat{Y}_i - \bar{Y})^2$ can be larger than $(Y_i - \bar{Y})^2$, overall we always have $ESS \leq TSS$.



3. (a) This follows directly from the first equation of the FOC for the OLS estimators:

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad (1)$$

i.e., $\sum_{i=1}^n \hat{u}_i = 0$.

- (b) **Solution 1:** Notice that $Y_i = \hat{Y}_i + \hat{u}_i$, i.e., the actual value of Y_i can be decomposed into the OLS predicted value $\hat{Y}_i$ and the residual $\hat{u}_i$. Therefore, we have

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i + \hat{u}_i) = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i + \frac{1}{n} \sum_{i=1}^n \hat{u}_i = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i, \quad (2)$$

where the last equality follows from the result that $\sum_{i=1}^n \hat{u}_i = 0$ in (a).

Solution 2: First, we have

$$\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}, \quad (3)$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Second, we notice that according to the formula for the OLS estimators, it holds that

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \quad (4)$$

Combining the two equations, we conclude that

$$\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}. \quad (5)$$

(c) **Solution 1:** If $\hat{\beta}_1 = 0$, then we have

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i = \hat{\beta}_0. \quad (6)$$

In addition, we have

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \bar{Y}, \quad (7)$$

where the first equality follows from the formula of OLS estimators. Combining these two equations, we obtain $\hat{Y}_i = \bar{Y}$. Therefore,

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n (\bar{Y} - \bar{Y})^2 = 0, \text{ and } R^2 = ESS/TSS = 0. \quad (8)$$

Solution 2: Notice that

$$\begin{aligned} ESS &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ &= \sum_{i=1}^n \left((\hat{\beta}_0 + \hat{\beta}_1 X_i) - (\hat{\beta}_0 + \hat{\beta}_1 \bar{X}) \right)^2 \\ &= \sum_{i=1}^n \left(\hat{\beta}_1 \cdot (X_i - \bar{X}) \right)^2 \end{aligned} \quad (9)$$

We can see that when $\hat{\beta}_1 = 0$, then $ESS = 0$.

4. (a) For 64 inches tall, $\widehat{Weight} = -79.24 + 4.16 \times 64 = 187$ pounds. In the same manner, for 68 and 72 inches, we obtain $\widehat{Weight} = 203.64$ pounds and 220.28 pounds, respectively.
- (b) $\Delta \widehat{Weight} = 4.16 \times \Delta Height = 4.16 \times 2 = 8.32$ (pounds).
- (c) We have the following relations: 1 inch = 2.54 cm, and 1 lb = 0.4536 kg. According to the formula in the last tutorial, if we multiply X by c_1 and Y by c_2 , where c_1 and c_2 are constants, then we have

$$\tilde{\beta}_1 = \left(\frac{c_2}{c_1} \right) \hat{\beta}_1, \quad \tilde{\beta}_0 = c_2 \hat{\beta}_0, \quad (10)$$

where $(\hat{\beta}_0, \hat{\beta}_1)$ denote the original OLS estimators computed from X and Y , while $(\tilde{\beta}_0, \tilde{\beta}_1)$ denote the new OLS estimators computed from $c_1 X$ and $c_2 Y$.

Furthermore, we can verify that if we multiply X by c_1 and Y by c_2 , the R^2 remains unchanged. The R^2 is in fact unit free.

Therefore, in the new centimeter-kilogram regression, the new slope estimate should be $(0.4536/2.54) \times 4.16 = 0.7429$, and the new intercept estimate should be $0.4536 \times (-79.24) = -35.9433$. On the other hand, the R^2 is still equal to 0.72.